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Chapter 4 - Separable First-Order Equations

This chapter turns the separable classification from Chapter 3 into a complete solution method.

A separable equation allows the dependence on the independent variable and the unknown function to be disentangled. Once that structure is exposed, the differential equation becomes an integration problem.

We will learn how to:

- recognise and separate the variables safely
- understand separation through the chain rule
- retain equilibrium solutions that division may remove
- solve initial-value problems with constants or definite integrals
- distinguish implicit and explicit answers
- choose the correct branch and interval of validity
- reduce first-order homogeneous equations to separable equations
- verify a result after all algebraic rearrangements are complete

The central warning is just as important as the method:

separation is powerful because it uses division, and division is exactly where solutions and domain restrictions can be lost.

Why Separation Works

The Goal Of This Block

The goal is to understand separation as a consequence of the chain rule, not as a mysterious movement of symbols.

This viewpoint makes the method easier to remember and safer to use.

The Separable Pattern

A first-order equation is separable when it can be written as:

$$y' = g(x)h(y)$$

Here:

- x is the independent variable
- $y = y(x)$ is the unknown function
- $g(x)$ depends only on x
- $h(y)$ depends only on y

Where $h(y) \neq 0$, divide by $h(y)$:

$$\frac{1}{h(y)} \frac{dy}{dx} = g(x)$$

Then write:

$$\frac{1}{h(y)} dy = g(x) dx$$

The left side depends only on y , and the right side depends only on x .

The Chain-Rule Explanation

Choose an antiderivative H satisfying:

$$H'(y) = \frac{1}{h(y)}$$

By the chain rule:

$$\frac{d}{dx}H(y(x)) = H'(y(x))y'(x)$$

Substitute $H'(y) = 1/h(y)$ into the chain-rule expression:

$$\frac{d}{dx}H(y(x)) = \frac{1}{h(y)}y'$$

Using the differential equation $y' = g(x)h(y)$ gives:

$$\frac{d}{dx}H(y(x)) = g(x)$$

If $G'(x) = g(x)$, integrate with respect to x :

$$H(y(x)) = G(x) + C$$

Equivalently:

$$\int \frac{1}{h(y)} dy = \int g(x) dx + C$$

The key intuition:

separation searches for a new quantity $H(y(x))$ whose derivative depends only on x .

The Differential-Form Version

A separable equation may instead be presented as:

$$A(x) dx + B(y) dy = 0$$

Integrate each one-variable term:

$$\int A(x) dx + \int B(y) dy = C$$

This form is already separated. No additional division is required.

The two common descriptions are related. From:

$$y' = g(x)h(y)$$

we can write, where $h(y) \neq 0$:

$$-g(x) dx + \frac{1}{h(y)} dy = 0$$

A First Worked Example

Solve:

$$y' = 2xe^{-y}$$

Step 1: Identify the product. The right side is:

$$g(x)h(y) = 2x e^{-y}$$

Step 2: Separate. Write $y' = dy/dx$ in the original equation:

$$\frac{dy}{dx} = 2xe^{-y}$$

Multiply both sides by $e^y dx$. On the right, $e^y e^{-y} = 1$, so:

$$e^y dy = 2x dx$$

Step 3: Integrate.

$$\int e^y dy = \int 2x dx$$

so:

$$e^y = x^2 + C$$

Step 4: Solve explicitly where possible. Take the natural logarithm:

$$y = \ln(x^2 + C)$$

This explicit form requires:

$$x^2 + C > 0$$

Step 5: Verify. Differentiate:

$$y' = \frac{2x}{x^2 + C}$$

From the proposed solution:

$$e^{-y} = \frac{1}{x^2 + C}$$

Therefore:

$$2xe^{-y} = \frac{2x}{x^2 + C} = y'$$

The equation is satisfied on every interval where $x^2 + C > 0$.

One Constant Is Enough

Suppose separate integration produces:

$$H(y) + C_1 = G(x) + C_2$$

Rearrange:

$$H(y) = G(x) + (C_2 - C_1)$$

The difference of two arbitrary constants is another arbitrary constant. Write:

$$C = C_2 - C_1$$

and obtain:

$$H(y) = G(x) + C$$

There is no need to carry two unrelated integration constants through the calculation.

Block 1 Summary

For:

$$y' = g(x)h(y)$$

the non-equilibrium part of the solution satisfies:

$$\int \frac{1}{h(y)} dy = \int g(x) dx + C$$

This formula follows from the chain rule. It is valid on regions where the division by $h(y)$ is permitted.

A Reliable Solution Workflow

The Goal Of This Block

The goal is to build a method that works consistently across straightforward and disguised separable equations.

The Seven-Step Method

Use this workflow.

Step 1: Isolate the derivative if useful. Try to expose:

$$y' = g(x)h(y)$$

Step 2: Find equilibrium candidates. Solve:

$$h(y) = 0$$

Each root may give a constant solution.

Step 3: State every division restriction. Before dividing by $h(y)$ or another expression, record where it vanishes.

Step 4: Separate the variables. Write one side using only y and dy , and the other using only x and dx .

Step 5: Integrate both sides. Use one arbitrary constant.

Step 6: Apply conditions and choose a branch. If an initial value is given, it determines the constant and may determine a sign or inverse-function branch.

Step 7: Verify and state the interval. Substitute into the original equation, not merely the separated equation. State an interval on which the formula and differential equation are both defined.

Example: A Solution That Stays Implicit

Solve:

$$(1 + y^2)y' = 2x - 1$$

Separate. Replace y' by dy/dx :

$$(1 + y^2) \frac{dy}{dx} = 2x - 1$$

Then multiply by dx :

$$(1 + y^2) dy = (2x - 1) dx$$

Integrate.

$$\int (1 + y^2) dy = \int (2x - 1) dx$$

Evaluating these two antiderivatives gives:

$$y + \frac{y^3}{3} = x^2 - x + C$$

The implicit solution is:

$$y + \frac{y^3}{3} = x^2 - x + C$$

There is no mathematical requirement to force this cubic relation into a long explicit formula.

In fact, the function:

$$L(y) = y + \frac{y^3}{3}$$

has derivative:

$$L'(y) = 1 + y^2 > 0$$

so L is strictly increasing and the implicit equation determines one real value of y for every real right-hand side.

What this tells us:

an implicit answer can describe the solution completely, even when solving algebraically for y would be inconvenient.

Verification By Implicit Differentiation

Start from:

$$y + \frac{y^3}{3} = x^2 - x + C$$

Differentiate both sides with respect to x :

$$y' + y^2 y' = 2x - 1$$

Factor y' :

$$(1 + y^2)y' = 2x - 1$$

This is the original equation.

The constant disappears under differentiation, as expected.

Example: Separation Hidden By A Quotient

Solve:

$$y' = \frac{1 + x^2}{2 + \cos y}$$

The denominator depends only on y , so multiply it across:

$$(2 + \cos y)y' = 1 + x^2$$

Now use $y' dx = dy$:

$$(2 + \cos y) dy = (1 + x^2) dx$$

Integrate:

$$\int (2 + \cos y) dy = \int (1 + x^2) dx$$

Evaluating both antiderivatives gives:

$$2y + \sin y = x + \frac{x^3}{3} + C$$

Thus:

$$2y + \sin y = x + \frac{x^3}{3} + C$$

Since:

$$2 + \cos y \geq 1$$

the original denominator never vanishes for real y . The left side is also strictly increasing as a function of y , so the implicit relation selects a unique real y for each real x .

A Nonseparable Near Miss

Consider:

$$y' = x^2 + y^2$$

The right side is a sum of an x -term and a y -term.

For the near miss $y' = x^2 + y^2$, writing:

$$\frac{dy}{x^2 + y^2} = dx$$

does not separate the variables because the left side still contains x and y .

By contrast:

$$y' = x^2 y^2$$

is separable because the right side is a product:

$$y^{-2} dy = x^2 dx$$

The practical test is not whether dy and dx appear on different sides. Each side must contain only its own variable.

Block 2 Summary

Separate, integrate, apply conditions, choose a branch, verify, and state the interval.

An implicit solution is often the natural final answer. Its validity can be checked directly by implicit differentiation.

Equilibrium Solutions And Lost Cases

The Goal Of This Block

The goal is to prevent division from silently removing constant solutions.

Find Zeros Before Dividing

For:

$$y' = g(x)h(y)$$

For the equation $y' = g(x)h(y)$, suppose a satisfies:

$$h(a) = 0$$

If $h(a) = 0$, consider the constant function:

$$y(x) = a$$

This function has:

$$y' = 0$$

For the constant function $y = a$, the right side is:

$$g(x)h(a) = 0$$

Therefore $y = a$ is a solution wherever the equation is defined.

But separation divides by $h(y)$, so this solution disappears from the separated equation.

The safe order is:

1. solve $h(y) = 0$
2. record the constant solutions
3. divide only for the remaining nonconstant branches

Worked Example With Two Equilibria

Solve:

$$y' = e^{-x}(y + 1)(3 - y)$$

Step 1: Find equilibrium solutions. The y -factor vanishes when:

$$y = -1 \quad \text{or} \quad y = 3$$

Thus two constant solutions are:

$$\boxed{y = -1} \quad \text{and} \quad \boxed{y = 3}$$

Step 2: Separate the nonconstant branches. For $y \neq -1, 3$:

$$\frac{1}{(y + 1)(3 - y)} dy = e^{-x} dx$$

Step 3: Decompose the left side. Write:

$$\frac{1}{(y + 1)(3 - y)} = \frac{A}{y + 1} + \frac{B}{3 - y}$$

Then:

$$1 = A(3 - y) + B(y + 1)$$

Set $y = -1$. The term containing B vanishes, leaving:

$$1 = A(4), \quad A = \frac{1}{4}$$

Set $y = 3$. The term containing A vanishes, leaving:

$$1 = B(4), \quad B = \frac{1}{4}$$

Substitute $A = B = 1/4$ into the partial-fraction decomposition:

$$\frac{1}{(y+1)(3-y)} = \frac{1}{4} \frac{1}{y+1} + \frac{1}{4} \frac{1}{3-y}$$

Step 4: Integrate carefully. Substitute the partial-fraction decomposition into the separated equation:

$$\left(\frac{1}{4} \frac{1}{y+1} + \frac{1}{4} \frac{1}{3-y} \right) dy = e^{-x} dx$$

Integrate both sides:

$$\frac{1}{4} \int \frac{1}{y+1} dy + \frac{1}{4} \int \frac{1}{3-y} dy = \int e^{-x} dx$$

Because:

$$\int \frac{1}{3-y} dy = -\ln|3-y|$$

and $\int e^{-x} dx = -e^{-x}$, we obtain:

$$\frac{1}{4} \ln|y+1| - \frac{1}{4} \ln|3-y| = -e^{-x} + C$$

Multiply every term by 4:

$$\ln|y+1| - \ln|3-y| = -4e^{-x} + C_1$$

Here $C_1 = 4C$ is still an arbitrary constant. Combine the logarithms, then rename C_1 as C :

$$\ln \left| \frac{y+1}{3-y} \right| = -4e^{-x} + C$$

This relation describes the nonconstant solutions. The two equilibrium solutions must be listed alongside it.

Can A Solution Cross An Equilibrium?

For many smooth equations, a solution cannot cross an equilibrium curve. For the equation just considered,

$$y' = e^{-x}(y+1)(3-y),$$

define the right-side function by:

$$f(x, y) = e^{-x}(y+1)(3-y)$$

This function has a continuous derivative with respect to y . Local uniqueness therefore holds.

If a nonconstant solution reached $y = 3$ at some point, it would share an initial value with the constant solution $y = 3$. Uniqueness would force the two solutions to agree locally, contradicting the supposed crossing.

So the horizontal equilibrium levels separate the solution family into regions.

equilibrium solutions are not merely discarded edge cases; they can form barriers that organise the behaviour of all other solutions.

When Uniqueness Fails

Separable equations can also reveal why hypotheses matter.

Consider:

$$y' = 3y^{2/3}$$

The constant function $y = 0$ is a solution.

For $y \neq 0$, separate:

$$y^{-2/3} dy = 3 dx$$

Integrating gives:

$$3y^{1/3} = 3x + C$$

Divide every term by 3:

$$y^{1/3} = x + \frac{C}{3}$$

Since $C/3$ is another arbitrary constant, write it as $-C_1$:

$$y^{1/3} = x - C_1$$

Cube both sides:

$$y = (x - C_1)^3$$

Now fix a number a and define:

$$y(x) = \begin{cases} 0, & x \leq a, \\ (x - a)^3, & x > a. \end{cases}$$

This function is differentiable at $x = a$ and satisfies the equation. It waits at the equilibrium and then leaves it.

The derivative of $3y^{2/3}$ with respect to y is not well behaved at $y = 0$, so the usual local uniqueness condition fails there.

The useful lesson is:

constant solutions must be checked directly, and uniqueness should not be assumed when the right side is not sufficiently regular in the dependent variable.

Block 3 Summary

Zeros of $h(y)$ produce equilibrium candidates for:

$$y' = g(x)h(y)$$

Record them before dividing by $h(y)$. Under ordinary uniqueness conditions, equilibrium curves cannot be crossed. Without those conditions, several solutions may pass through the same initial point.

Initial-Value Problems

The Goal Of This Block

The goal is to use an initial condition efficiently and to understand how it selects one member of a solution family.

The Constant-Then-Condition Method

For an initial-value problem:

$$y' = g(x)h(y), \quad y(x_0) = y_0$$

one approach is:

1. separate and integrate
2. obtain $H(y) = G(x) + C$
3. substitute (x_0, y_0)
4. solve for C

This is familiar and usually effective.

The Definite-Integral Form

There is an alternative that builds the initial condition into the equation.

Assume the solution remains in a region where $h(y) \neq 0$. From:

$$\frac{1}{h(y)} dy = g(x) dx$$

integrate from the initial point to the current point:

$$\boxed{\int_{y_0}^{y(x)} \frac{1}{h(u)} du = \int_{x_0}^x g(s) ds}$$

The symbols u and s are dummy integration variables. They prevent the upper limits $y(x)$ and x from being confused with the variables inside the integrals.

At $x = x_0$, both integrals are zero, so the initial condition is automatically encoded.

Worked Initial-Value Problem

Solve:

$$y' = \frac{1 + x^2}{2 + \cos y}, \quad y(0) = \frac{\pi}{2}$$

Multiply the equation by $(2 + \cos y) dx$. Since this factor is always positive, the operation introduces no excluded real values:

$$(2 + \cos y) dy = (1 + x^2) dx$$

Use definite integrals:

$$\int_{\pi/2}^{y(x)} (2 + \cos u) du = \int_0^x (1 + s^2) ds$$

Evaluate the left side:

$$\begin{aligned} \int_{\pi/2}^y (2 + \cos u) du &= [2u + \sin u]_{\pi/2}^y \\ &= 2y + \sin y - (\pi + 1) \end{aligned}$$

Evaluate the right side:

$$\int_0^x (1 + s^2) ds = x + \frac{x^3}{3}$$

Therefore the solution is determined implicitly by:

$$\boxed{2y + \sin y - (\pi + 1) = x + \frac{x^3}{3}}$$

Check the initial condition by setting $x = 0$ and $y = \pi/2$. Both sides are zero.

Because $2 + \cos y$ is always positive, the left side is strictly increasing in y . The implicit relation therefore selects one real branch without any sign ambiguity.

An Initial Value At An Equilibrium

Consider:

$$y' = x^2y(2 - y), \quad y(1) = 0$$

If we immediately divide by $y(2 - y)$, the initial value $y = 0$ is excluded. Instead, test the equilibrium first.

For:

$$y(x) = 0$$

we have:

$$y' = 0$$

and:

$$x^2y(2 - y) = 0$$

The initial condition is also satisfied. Thus:

$$\boxed{y(x) = 0}$$

is the solution of the initial-value problem. Because the right side is smooth in y , local uniqueness rules out a second branch through $(1, 0)$.

Branch Selection

An implicit relation can represent more than one branch.

Suppose integration gives:

$$(y - 1)^2 = R(x)$$

Solving explicitly gives:

$$y = 1 + \sqrt{R(x)}$$

or:

$$y = 1 - \sqrt{R(x)}$$

An initial condition determines which sign is compatible with y_0 .

The sign is not a free choice at every point. A differentiable solution must remain on a consistent branch until the equation or uniqueness conditions allow otherwise.

Block 4 Summary

An initial condition may be applied after indefinite integration or built in through definite integrals:

$$\int_{y_0}^{y(x)} \frac{du}{h(u)} = \int_{x_0}^x g(s) ds$$

Check equilibrium initial values before dividing. Use the initial value to select constants, signs, and branches.

Domains, Singularities, And Maximal Intervals

The Goal Of This Block

The goal is to treat a solution as a formula together with an interval on which that formula actually solves the equation.

Why A Formula Is Not The Whole Answer

A proposed solution may fail where:

- the original differential equation is undefined
- a denominator in the solution vanishes
- a logarithm receives a nonpositive argument
- an even root receives a negative argument
- an inverse-function branch changes
- the solution becomes unbounded

For an initial-value problem, we want the largest connected interval containing the initial point on which the solution remains valid. This is the **maximal interval of existence** for that branch.

Worked Example: Finite-Time Blow-Up

Solve:

$$u' = 3su^2, \quad u(0) = 2$$

Check the equilibrium. The value $u = 0$ is a constant solution, but it does not satisfy the initial condition.

Separate the nonzero branch.

$$u^{-2} du = 3s ds$$

Integrate.

$$-\frac{1}{u} = \frac{3}{2}s^2 + C$$

Apply the initial condition. At $s = 0$ and $u = 2$:

$$-\frac{1}{2} = C$$

Substitute $C = -1/2$ into the integrated relation:

$$-\frac{1}{u} = \frac{3}{2}s^2 - \frac{1}{2}$$

Solve for u :

$$-\frac{1}{u} = \frac{3s^2 - 1}{2}$$

Multiply by -1 :

$$\frac{1}{u} = \frac{1 - 3s^2}{2}$$

Take reciprocals where $1 - 3s^2 \neq 0$:

$$u(s) = \frac{2}{1 - 3s^2}$$

Find the maximal interval. The denominator vanishes when:

$$1 - 3s^2 = 0$$

so:

$$s = \pm \frac{1}{\sqrt{3}}$$

The largest interval containing the initial point $s = 0$ is:

$$\boxed{-\frac{1}{\sqrt{3}} < s < \frac{1}{\sqrt{3}}}$$

The differential equation itself is smooth everywhere, but this particular solution becomes unbounded at finite values of s .

Logarithms Carry Interval Information

When integration produces:

$$\ln |y - a| = G(x) + C$$

the absolute value indicates two possible regions:

$$y > a$$

or:

$$y < a$$

Exponentiating gives:

$$|y - a| = e^{G(x)+C} = e^C e^{G(x)}$$

Set $K = e^C$, so $K > 0$:

$$|y - a| = K e^{G(x)}$$

The absolute value permits either sign. Absorb that sign into a nonzero constant C_1 :

$$y - a = C_1 e^{G(x)}$$

The case $C_1 = 0$ corresponds to the equilibrium $y = a$ if that solution is valid. Whether it was present in the separated derivation must be checked against the original equation.

Inverse Trigonometric Functions Need Branch Care

Consider:

$$q' = (1 + q^2) \cos x, \quad q(0) = 0$$

Divide by $1 + q^2$ and use $q' dx = dq$ to separate:

$$\frac{1}{1 + q^2} dq = \cos x dx$$

Integrate:

$$\arctan q = \sin x + C$$

Apply the initial condition to the implicit relation:

$$\arctan(0) = \sin(0) + C, \quad 0 = 0 + C, \quad C = 0$$

Substitute $C = 0$ and apply tangent to both sides:

$$\boxed{q(x) = \tan(\sin x)}$$

For the proposed solution $q(x) = \tan(\sin x)$, since:

$$-1 \leq \sin x \leq 1$$

and:

$$[-1, 1] \subset \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

the tangent never reaches a pole. This solution is defined for every real x . The interval conclusion comes from examining the range of the inner function, not merely from writing down a tangent.

Implicit Solutions Also Have Domains

Suppose a solution is left as:

$$F(x, y) = C$$

Implicit differentiation gives:

$$F_x + F_y y' = 0$$

so:

$$y' = -\frac{F_x}{F_y}$$

The curve can be represented locally as a differentiable function $y(x)$ only where:

$$F_y \neq 0$$

Thus an implicit formula does not eliminate domain questions. It changes the way those questions are inspected.

Block 5 Summary

A solution consists of a formula or implicit relation together with a valid interval.

For an initial-value problem, choose the largest connected interval containing the initial point that avoids singularities, invalid branches, and points where the solution ceases to satisfy the original equation.

First-Order Homogeneous Equations Become Separable

The Goal Of This Block

The goal is to solve the scale-homogeneous equations classified in Chapter 3 by converting them into separable equations.

The Substitution

Suppose:

$$y' = H\left(\frac{y}{x}\right)$$

on a region where $x \neq 0$.

For $y' = H(y/x)$, set:

$$v = \frac{y}{x}$$

so:

$$y = xv$$

Because $v = v(x)$ and $y = xv$, differentiate with the product rule:

$$y' = v + xv'$$

Substitute into the equation:

$$v + xv' = H(v)$$

Subtract v from both sides:

$$xv' = H(v) - v$$

Where $H(v) - v \neq 0$, separate:

$$\boxed{\frac{1}{H(v) - v} dv = \frac{1}{x} dx}$$

After integration, replace v by y/x .

Special Ratio Solutions

Before dividing by $H(v) - v$, solve:

$$H(v) = v$$

If $v = m$ is a root of $H(v) = v$, then $v' = 0$ and:

$$v = \frac{y}{x} = m$$

Since $v = y/x = m$, multiply by x to obtain the straight-line candidate:

$$\boxed{y = mx}$$

These are the homogeneous-equation analogue of equilibrium solutions. They can be lost during separation in the transformed variable.

Worked Homogeneous Example

Solve:

$$y' = \frac{y}{x} + 1 + \left(\frac{y}{x}\right)^2$$

The slope depends only on y/x , so use:

$$v = \frac{y}{x}, \quad y' = v + xv'$$

Substitute:

$$v + xv' = v + 1 + v^2$$

Subtract v from both sides:

$$xv' = 1 + v^2$$

Starting from $xv' = 1 + v^2$, divide by $1 + v^2$ and by x to separate:

$$\frac{1}{1 + v^2} dv = \frac{1}{x} dx$$

Integrate the separated equation:

$$\int \frac{1}{1+v^2} dv = \int \frac{1}{x} dx$$

Evaluating the antiderivatives gives:

$$\arctan v = \ln |x| + C$$

Apply tangent to both sides on a branch where the expression is defined:

$$v = \tan(\ln |x| + C)$$

Back-substitute $v = y/x$:

$$\frac{y}{x} = \tan(\ln |x| + C)$$

Multiply by x :

$$\boxed{y = x \tan(\ln |x| + C)}$$

This family is valid on intervals that:

- do not cross $x = 0$
- do not contain a point where $\ln |x| + C$ is an odd multiple of $\pi/2$

There are no real special-ratio solutions here because:

$$1 + v^2 \neq 0$$

for real v .

Homogeneous Differential Form

Consider:

$$M(x, y) dx + N(x, y) dy = 0$$

Suppose M and N are homogeneous functions of the same degree k :

$$M(\lambda x, \lambda y) = \lambda^k M(x, y)$$

and:

$$N(\lambda x, \lambda y) = \lambda^k N(x, y)$$

Where $N \neq 0$, the slope is:

$$y' = -\frac{M(x, y)}{N(x, y)}$$

Under common scaling, the factors λ^k cancel:

$$-\frac{M(\lambda x, \lambda y)}{N(\lambda x, \lambda y)} = -\frac{M(x, y)}{N(x, y)}$$

Thus the slope is degree zero and can be expressed as a function of y/x on an appropriate region.

Block 6 Summary

For:

$$y' = H(y/x)$$

For $y' = H(y/x)$, use:

$$y = xv, \quad y' = v + xv'$$

The transformed equation is separable. Check $H(v) = v$ before dividing, then back-substitute $v = y/x$ and respect the restriction $x \neq 0$.

Choosing The Reciprocal Homogeneous Substitution

The Goal Of This Block

The goal is to recognise when treating x as a function of y makes a homogeneous equation simpler.

The Reciprocal View

If $y' \neq 0$, then locally:

$$\frac{dx}{dy} = \frac{1}{dy/dx}$$

For an equation whose reciprocal slope depends naturally on x/y , set:

$$u = \frac{x}{y}$$

so:

$$x = yu$$

Differentiate with respect to y :

$$\frac{dx}{dy} = u + y \frac{du}{dy}$$

The resulting equation may separate more cleanly than the $v = y/x$ substitution.

This is an alternative route, not a new class of equation.

Worked Reciprocal Example

Solve:

$$\frac{dy}{dx} = \frac{y}{2y - x}$$

On a branch where taking the reciprocal is valid, invert the original slope:

$$\frac{dx}{dy} = \frac{2y - x}{y} = 2 - \frac{x}{y}$$

Set:

$$u = \frac{x}{y}, \quad x = yu$$

Then:

$$\frac{dx}{dy} = u + y \frac{du}{dy}$$

Substitute:

$$u + y \frac{du}{dy} = 2 - u$$

Therefore:

$$y \frac{du}{dy} = 2(1 - u)$$

For $u \neq 1$, separate:

$$\frac{1}{1-u} du = 2 \frac{1}{y} dy$$

Integrate the separated equation:

$$\int \frac{1}{1-u} du = 2 \int \frac{1}{y} dy$$

Evaluating the antiderivatives gives:

$$-\ln|1-u| = 2 \ln|y| + C$$

Starting from $-\ln|1-u| = 2 \ln|y| + C$, move the logarithmic term involving y to the left and absorb the changed sign of the arbitrary constant into C_1 :

$$\ln|1-u| + 2 \ln|y| = C_1$$

Because $2 \ln|y| = \ln(y^2)$ for $y \neq 0$, combine the logarithms:

$$\ln|(1-u)y^2| = C_1$$

Exponentiate:

$$|(1-u)y^2| = e^{C_1}$$

Absorb the possible sign and the positive factor e^{C_1} into one nonzero constant C . This gives:

$$(1-u)y^2 = C$$

Back-substitute $u = x/y$:

$$\left(1 - \frac{x}{y}\right)y^2 = C$$

Distribute y^2 using $y \neq 0$ on the transformed branch:

$$y^2 - \frac{x}{y}y^2 = C$$

Since $(x/y)y^2 = xy$, this becomes:

$$\boxed{y^2 - xy = C}$$

Check the excluded cases. The substitution $u = x/y$ assumed $y \neq 0$. In the original equation, $y = 0$ gives $y' = 0$ wherever $x \neq 0$, so:

$$\boxed{y = 0}$$

is a solution on intervals that do not contain the origin.

The value $u = 1$ corresponds to:

$$x = y$$

or $y = x$. This is also a solution wherever $x \neq 0$.

When $C = 0$, the implicit relation factors as:

$$y(y - x) = 0$$

so it contains both lines. They should be understood as separate solution branches rather than one branch switching lines at the origin, where the original equation is undefined.

Verify. Differentiate:

$$2yy' - (y + xy') = 0$$

Thus:

$$(2y - x)y' = y$$

and:

$$y' = \frac{y}{2y - x}$$

where the denominator is nonzero.

How To Choose

Prefer $v = y/x$ when the original slope is already simple as a function of y/x .

Consider $u = x/y$ when:

- the reciprocal slope is simpler
- the equation naturally contains repeated x/y expressions
- solving for dx/dy removes a complicated quotient

Both substitutions encode the same scale-invariant geometry. The best choice is the one that makes the transformed equation easiest to separate and integrate.

Block 7 Summary

The standard homogeneous substitution is $y = xv$, but the reciprocal form $x = yu$ may be more efficient.

Whichever route is used, differentiate the product correctly, check any divided cases, back-substitute, and verify in the original equation.

A Complete Problem-Solving Checklist

Before Calculating

Ask:

1. Is the equation truly separable?
2. Which factors depend only on the independent variable?
3. Which factors depend only on the unknown function?
4. Which expressions could become zero?
5. Are there equilibrium or special-ratio solutions?

During The Calculation

Check that:

1. each side contains only one variable after separation
2. every division restriction is recorded
3. antiderivatives include the correct signs and absolute values
4. only one arbitrary constant is retained
5. substitutions are differentiated with the chain or product rule

After Integrating

Decide:

1. whether implicit form is already complete
2. whether solving explicitly introduces several branches
3. which branch satisfies the initial condition
4. where the formula and original equation are defined
5. what the maximal interval containing the initial point is
6. whether all excluded constant or ratio solutions have been restored
7. whether differentiation reproduces the original equation

When Exact Algebra Is Not Available

Separation may lead to an integral with no elementary antiderivative:

$$\int \frac{1}{h(y)} dy = \int g(x) dx + C$$

This does not make the method useless.

The integral relation may still:

- define the solution implicitly
- support numerical quadrature
- reveal monotonicity
- identify equilibria and barriers
- provide a starting point for approximation

Similarly, an antiderivative may be available while algebraic inversion is not. Exact mathematics does not require every final answer to be written as $y =$ followed by elementary functions.

Block 8 Summary

The separation method is more than integrating two expressions. A complete solution accounts for excluded cases, constants, branches, domains, and verification.

the safest separable solution is the one whose algebra, missing cases, and interval have all been checked against the original equation.

Original Practice Set

Practice Problems

Problem 1: Recognise The Structure. Determine which equations are separable as written or after straightforward algebra.

- (a) $y' = \frac{1 + x^2}{2 + y^2},$
- (b) $y' = x + y,$
- (c) $y' = x(y + 1),$
- (d) $y' = \sin(xy).$

Problem 2: Find An Implicit Family. Solve:

$$(1 + y^2)y' = 4x$$

and leave the answer in a natural implicit form.

Problem 3: Find An Explicit Family. Solve:

$$y' = -2te^{-y}$$

and state the restriction required by the explicit logarithmic form.

Problem 4: Retain Every Equilibrium. For:

$$y' = (1 + x^2)y(y + 4)$$

identify all constant solutions and write the separated equation for the remaining branches. You do not need to complete the partial-fraction integration.

Problem 5: Solve An Initial-Value Problem Implicitly. Solve:

$$(1 + q^2)q' = e^x, \quad q(0) = 0$$

in implicit form.

Problem 6: Use Definite Integrals. Write a definite-integral equation for the solution of:

$$r' = (2 + t^2)(1 + r^2), \quad r(1) = -1$$

Do not evaluate the integrals.

Problem 7: Find A Maximal Interval. Solve:

$$u' = 2xu^2, \quad u(0) = 1$$

and find the maximal interval containing 0.

Problem 8: Verify An Implicit Solution. Show by implicit differentiation that the relation $e^y + y = x^3 + C$ satisfies the differential equation:

$$(e^y + 1)y' = 3x^2$$

Problem 9: Detect A Homogeneous Equation. Determine whether:

$$y' = 2 + \frac{y}{x} - \left(\frac{y}{x}\right)^2$$

is first-order homogeneous, and state the substitution that reduces it to a separable equation.

Problem 10: Solve A Homogeneous Equation. Solve:

$$y' = \frac{y}{x} + \left(\frac{y}{x}\right)^2$$

including any special-ratio solution lost during separation.

Problem 11: Use The Reciprocal Substitution. For:

$$\frac{dy}{dx} = \frac{y}{3y - 2x'}$$

rewrite the equation using $u = x/y$ and obtain the transformed differential equation for $u(y)$. You do not need to integrate it.

Problem 12: Diagnose An Incomplete Answer. For $y' = x(y-2)$, a calculation ends with:

$$\ln |y - 2| = \frac{x^2}{2} + C$$

What solution has been omitted? How can the nonconstant family be rewritten so that the omitted solution is naturally restored?

Practice Answers

Answer 1. Equation (a) is separable:

$$(2 + y^2) dy = (1 + x^2) dx$$

Equation (b) is not separable by the standard test because $x + y$ is an additive mixture of the variables.

Equation (c) is separable:

$$\frac{1}{y+1} dy = x dx$$

for $y \neq -1$. The constant solution $y = -1$ must be checked separately.

For $y = -1$, the left side is $y' = 0$, while the original right side is $x(-1 + 1) = 0$. Hence $y = -1$ is indeed a solution.

Equation (d) is not ordinarily separable because x and y occur together inside one sine function.

Answer 2. Replace y' by dy/dx in the given equation:

$$(1 + y^2) \frac{dy}{dx} = 4x$$

Starting from $(1 + y^2)dy/dx = 4x$, multiply by dx to separate:

$$(1 + y^2) dy = 4x dx$$

Integrate:

$$\int (1 + y^2) dy = \int 4x dx$$

Evaluating $\int (1 + y^2) dy$ and $\int 4x dx$ gives:

$$y + \frac{y^3}{3} = 2x^2 + C$$

Answer 3. Starting from $y' = -2te^{-y}$, write $y' = dy/dt$ and multiply by $e^y dt$:

$$e^y dy = -2t dt$$

Integrate:

$$\int e^y dy = \int -2t dt$$

Therefore:

$$e^y = -t^2 + C$$

Take the natural logarithm of both sides. This requires $C - t^2 > 0$ and gives:

$$y = \ln(C - t^2)$$

The logarithmic formula is valid on any interval satisfying:

$$C - t^2 > 0$$

Answer 4. The roots of the y -factor are:

$$y = 0 \quad \text{and} \quad y = -4$$

Both give constant solutions.

Indeed, for either $y = 0$ or $y = -4$, we have $y' = 0$ and $(1 + x^2)y(y + 4) = 0$, so the original equation becomes $0 = 0$.

For the remaining branches, $y \neq 0, -4$. Divide the original equation by $y(y + 4)$ and multiply by dx :

$$\frac{1}{y(y + 4)} dy = (1 + x^2) dx$$

with $y \neq 0, -4$.

Answer 5. Replace q' by dq/dx and multiply by dx :

$$(1 + q^2) dq = e^x dx$$

Integrate:

$$\int (1 + q^2) dq = \int e^x dx$$

Evaluating the two antiderivatives gives:

$$q + \frac{q^3}{3} = e^x + C$$

Apply $q(0) = 0$ to $q + q^3/3 = e^x + C$:

$$0 + \frac{0^3}{3} = e^0 + C = 1 + C$$

Subtract 1 from both sides to obtain $C = -1$. Substituting this value into the implicit family gives:

$$q + \frac{q^3}{3} = e^x - 1$$

Answer 6. Write $r' = dr/dt$ and divide by $1 + r^2$. This division is always valid for real r because $1 + r^2 > 0$:

$$\frac{1}{1 + r^2} dr = (2 + t^2) dt$$

The definite-integral equation is:

$$\int_{-1}^{r(t)} \frac{1}{1 + v^2} dv = \int_1^t (2 + s^2) ds$$

Answer 7. The equilibrium $u = 0$ does not satisfy the initial condition. For $u \neq 0$:

$$u^{-2} du = 2x dx$$

Integrate the separated equation:

$$\int u^{-2} du = \int 2x dx$$

Evaluating $\int u^{-2} du$ and $\int 2x dx$ gives:

$$-\frac{1}{u} = x^2 + C$$

Use $u(0) = 1$:

$$-\frac{1}{1} = 0^2 + C, \quad C = -1$$

Substitute $C = -1$ into the implicit relation:

$$-\frac{1}{u} = x^2 - 1$$

Starting from $-1/u = x^2 - 1$, multiply by -1 and take reciprocals where $1 - x^2 \neq 0$:

$$\frac{1}{u} = 1 - x^2$$

Therefore:

$$u = \frac{1}{1 - x^2}$$

The singularities are at $x = \pm 1$, so the maximal interval containing 0 is:

$$-1 < x < 1$$

Answer 8. Differentiate:

$$\frac{d}{dx}(e^y + y) = e^y y' + y'$$

and:

$$\frac{d}{dx}(x^3 + C) = 3x^2$$

Therefore:

$$(e^y + 1)y' = 3x^2$$

which is the required equation.

Answer 9. The right side depends only on y/x :

$$H(v) = 2 + v - v^2$$

so the equation is first-order homogeneous on a region with $x \neq 0$.

Use:

$$v = \frac{y}{x}, \quad y = xv$$

with:

$$y' = v + xv'$$

Substitute $y' = v + xv'$ and $y/x = v$ into the original equation:

$$v + xv' = 2 + v - v^2$$

Subtract v from both sides:

$$xv' = 2 - v^2$$

which is separable.

Answer 10. Use:

$$v = \frac{y}{x}, \quad y' = v + xv'$$

Then:

$$v + xv' = v + v^2$$

Starting from $v + xv' = v + v^2$, subtract v from both sides:

$$xv' = v^2$$

The special value $v = 0$ gives:

$$\frac{y}{x} = 0$$

The special value $v = 0$ gave $y/x = 0$. Since $x \neq 0$ on the region under consideration, this means:

$$\boxed{y = 0}$$

For $v \neq 0$:

$$v^{-2} dv = \frac{1}{x} dx$$

Integrate:

$$\int v^{-2} dv = \int \frac{1}{x} dx$$

Hence:

$$-\frac{1}{v} = \ln |x| + C$$

Multiply by -1 and take reciprocals, requiring $\ln |x| + C \neq 0$:

$$v = -\frac{1}{\ln |x| + C}$$

Now replace v by y/x :

$$\frac{y}{x} = -\frac{1}{\ln |x| + C}$$

Starting from $y/x = -1/(\ln |x| + C)$, multiply both sides by x :

$$y = -\frac{x}{\ln |x| + C}$$

on intervals where $x \neq 0$ and $\ln |x| + C \neq 0$.

Answer 11. Take the reciprocal:

$$\frac{dx}{dy} = 3 - 2\frac{x}{y}$$

This reciprocal form is used on a branch where the required denominators are nonzero. In particular, it excludes the line $y = 0$ during this calculation. Direct substitution into the original equation shows that $y = 0$ is also a solution on intervals where $x \neq 0$. The problem asks only for the transformed nonzero branch.

Set:

$$u = \frac{x}{y}, \quad x = yu$$

Then:

$$\frac{dx}{dy} = u + y \frac{du}{dy}$$

so:

$$u + y \frac{du}{dy} = 3 - 2u$$

Therefore:

$$\boxed{y \frac{du}{dy} = 3(1 - u)}$$

which is separable.

Answer 12. Division by $y - 2$ omitted the equilibrium solution:

$$\boxed{y = 2}$$

Start from the given nonconstant relation:

$$\ln |y - 2| = \frac{x^2}{2} + C$$

Exponentiate both sides:

$$|y - 2| = e^{x^2/2+C} = e^C e^{x^2/2}$$

Set $K = e^C > 0$:

$$|y - 2| = K e^{x^2/2}$$

Absorb the sign permitted by the absolute value into a nonzero constant A :

$$y - 2 = A e^{x^2/2}$$

so the nonconstant family is:

$$y = 2 + A e^{x^2/2}$$

Allowing $A = 0$ restores the equilibrium, giving the unified family:

$$\boxed{y = 2 + A e^{x^2/2}}$$

for arbitrary real A .

Final Takeaway

Separable equations turn a differential relationship into two one-variable integrals.

The method is conceptually simple:

$$y' = g(x)h(y) \implies \frac{dy}{h(y)} = g(x) dx$$

but a complete solution requires more than integration.

We must preserve equilibria, apply conditions, choose branches, respect domains, identify maximal intervals, and verify the result in the original equation.

The main lesson is:

separate the variables, but keep the full solution problem connected.